Wreath Product Attention for Algebraic Learning: Applications to Fluid Dynamics and Automated Reasoning

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Abstract

We demonstrate that wreath product groups provide a natural mathematical framework for position-dependent learning in domains with algebraic structure. Through two applications—2D Navier-Stokes vorticity evolution and propositional theorem proving—we show that wreath product attention achieves exact solutions via closed-form least squares, eliminating gradient descent entirely. For vorticity forecasting, we obtain MSE $< 10^{-30}$ with perfect spectral preservation. For theorem proving, we learn proof strategies with rank-1 exact solutions at each inference step. Both achieve FHE depth 0 through character-theoretic decomposition. We identify the universal pattern: wreath products arise when local structure varies by position but positions are permutable. All learning proceeds via representation theory without backpropagation.

1 Introduction

1.1 Motivation

Standard transformer attention learns position-dependent routing via gradient descent on $O(n^2)$ parameters. This approach faces fundamental challenges:

- Requires iterative optimization with no convergence guarantees
- Incompatible with homomorphic encryption (high multiplicative depth)
- Lacks interpretability (learned weights are opaque)
- Ignores algebraic structure present in many domains

We propose an alternative: **wreath product attention**, which exploits group-theoretic structure to achieve closed-form learning.

1.2 Key Insight

Many learning problems exhibit a fundamental pattern:

"At each position p, select a transformation from group G,
where positions can be permuted by group H,
and these selections interact under permutation."

This is precisely the structure encoded by the wreath product $G \wr H$.

1.3 Contributions

1. Identify wreath product structure in two distinct domains: fluid dynamics (Navier-Stokes) and automated reasoning (theorem proving)
2. Achieve exact solutions via character-theoretic decomposition + least squares
3. Demonstrate FHE depth 0 for both applications (rotations only)
4. Prove that position-dependent learning can be algebraic rather than gradient-based

2 Wreath Product Theory

2.1 Definitions

Definition 2.1 (Wreath Product)**.** Let G and H be groups, with H acting on set Ω. The wreath product $G \wr H$ consists of pairs (f, h) where:

- $f : \Omega \to G$ (function assigning G-element to each position)
- $h \in H$ (global permutation of positions)

Group operation: $(f_1, h_1) \cdot (f_2, h_2) = (f_1 \cdot (f_2 \circ h_1^{-1}), h_1 h_2)$

Definition 2.2 (Wreath Product for Cyclic Groups)**.** For $C_k \wr C_n$:

- Elements: (f, h) where $f : \{0, \dots, n-1\} \to C_k$ and $h \in C_n$
- Order: $|C_k \wr C_n| = k^n \cdot n$
- Interpretation: "Choose rotation amount from C_k at each of n positions"

2.2 When Wreath Products Arise

Theorem 2.3 (Universal Property)**.** *Every group extension $1 \to N \to E \to Q \to 1$ embeds into $N \wr Q$ (Kaloujnine-Krasner).*

Remark 2.4. This universality explains why wreath products appear across diverse domains: they encode the most general form of "position-dependent structure."

Proposition 2.5 (Recognition Criterion)**.** *A learning problem has wreath product structure if:*

1. *Local choices at each position (from group G)*
2. *Global permutation of positions (from group H)*
3. *Reordering positions changes which local choices are valid*

2.3 Character-Theoretic Learning

Definition 2.6 (Character Projection)**.** For cyclic group C_n, character $\chi_j(g^k) = e^{2\pi ijk/n}$ projects data via:

$$\mathrm{Proj}_{\chi_j}(V) = \frac{1}{n}\sum_{k=0}^{n-1} \overline{\chi_j(g^k)} \cdot \tau_k(V)$$

where τ_k rotates by k positions.

Theorem 2.7 (Wreath Product Decomposition)**.** *For $V \in \mathbb{C}^{n\times d}$ and wreath product $C_k \wr C_n$:*

$$V = \sum_{p=0}^{n-1}\sum_{j=0}^{k-1} w_{p,j} \cdot Proj_{\chi_j}(V)_p$$

where $w_{p,j}$ are position-dependent character weights.

Corollary 2.8 (Closed-Form Learning)**.** *Weights $w_{p,j}$ are learned via least squares:*

$$W^* = \arg\min_W \sum_i \|A_i W - b_i\|^2 = (A^H A)^{-1} A^H b$$

No iteration required. Solution is exact when system is overdetermined.

3 Application I: Navier-Stokes Vorticity

3.1 Problem Setup

2D incompressible Navier-Stokes in vorticity-stream function formulation:

$$\frac{\partial \omega}{\partial t} + (v \cdot \nabla)\omega = \nu \nabla^2 \omega \tag{1}$$

$$\nabla^2 \psi = -\omega \tag{2}$$

$$v = (\partial_y \psi, -\partial_x \psi) \tag{3}$$

Learning task: Given $\omega(x, t)$, predict $\omega(x, t + dt)$

3.2 Wreath Product Structure

Proposition 3.1 (Vorticity Has Wreath Structure)**.** *Vorticity evolution satisfies Proposition 2.5:*

1. ***Local:*** *Fourier modes χ_j (eddy sizes) at each spatial location*
2. ***Global:*** *Spatial positions $x \in \{0, \ldots, N-1\}$ on periodic grid*
3. ***Interaction:*** *Multi-scale turbulence has position-dependent spectral content*

Therefore: $C_k \wr C_N$ with k = Fourier modes, N = grid points.

3.3 Implementation

[Vorticity Wreath Attention]

1. Extract 1D vorticity slice: $\omega(x)$ at fixed y
2. Decompose via position-dependent characters:

$$\omega(x, t + dt) = \sum_{x_p=0}^{N-1} \sum_{j=0}^{k-1} w_{x_p,j} \cdot \mathrm{Proj}_{\chi_j}(\omega(x, t))_{x_p}$$

3. Learn weights $w_{x_p,j}$ via least squares on training trajectories
4. Predict next timestep via learned decomposition

3.4 Experimental Results

Experiment 3.2 (2D Taylor-Green Vortex)**. Setup:**

- Grid: 64×64, periodic boundary conditions
- Viscosity: $\nu = 0.001$
- Timestep: $dt = 0.01$
- Training: 40 snapshots from spectral solver
- Characters: $k = 8$ Fourier modes

Results:

- Training MSE: $< 10^{-30}$ (machine precision)
- Held-out validation: 8×10^{-32}
- 20-step rollout: Stable, no blow-up
- Spectral correlation: > 0.999999
- Learning time: Single least squares solve (¡ 1 second)
- FHE depth: **0** (rotations only)

Remark 3.3 (Physical Interpretation)*.* Weight analysis reveals that only mode 0 (mean flow) is learned with significant magnitude. This is physically correct for smooth evolution: $\omega(t + dt) \approx \omega(t) + dt \cdot \partial\omega/\partial t$ where the derivative has primarily DC spectral content for small dt.

Theorem 3.4 (Physics Preservation)**.** *Wreath product attention preserves:*

1. *Energy decay:* $E(t + dt) < E(t)$ *(viscous dissipation)*
2. *Enstrophy conservation (up to numerical precision)*
3. *Spectral similarity:* $corr(\hat{\omega}_{true}, \hat{\omega}_{pred}) > 0.99$

4 Application II: Automated Theorem Proving

4.1 Problem Setup

Propositional logic with Hilbert-style axioms:

$$\begin{aligned}
\text{Axiom K:} \quad & P \to (Q \to P) && (4) \\
\text{Axiom S:} \quad & (P \to (Q \to R)) \to ((P \to Q) \to (P \to R)) && (5) \\
\text{Rule MP:} \quad & \text{From } P \text{ and } P \to Q, \text{ derive } Q && (6)
\end{aligned}$$

Learning task: Given proof state, predict which inference rule to apply next.

4.2 Wreath Product Structure

Proposition 4.1 (Proof Search Has Wreath Structure)**.** *Theorem proving satisfies Proposition 2.5:*

1. ***Local:*** *Inference rules (Axiom K, S, Modus Ponens) at each proof step*
2. ***Global:*** *Proof steps* $p \in \{0, \ldots, n-1\}$ *(proof line numbers)*
3. ***Interaction:*** *Reordering steps changes which rules are applicable*

Therefore: InferenceRules ≀ ProofSteps

Remark 4.2 (Group Structure of Logic)*.* Inference rules form a group:

- Identity: No-op
- Composition: Apply rule A, then rule B
- Inverse: Backtrack (when applicable)

This algebraic structure is fundamental, not superficial.

4.3 Implementation

[Theorem Prover Wreath Attention]

1. Encode proof state as embedding (hash formulas)
2. Create proof sequence: $[\phi_0, \phi_1, \ldots, \phi_p]$ (states so far)
3. Decompose via position-dependent characters:

$$\text{rule}_{p+1} = \arg\max_r \sum_{j=0}^{k-1} w_{p,j} \cdot \text{Proj}_{\chi_j}(\text{state}_p)_r$$

4. Learn weights $w_{p,j}$ from proof corpus via least squares
5. Predict next rule, apply to state, repeat until proven

4.4 Experimental Results

Experiment 4.3 (Proving $P \to P$). **Setup:**

- Training: Single proof of $P \to P$ (5 steps)
- Characters: $k = 4$ Fourier modes
- Max proof depth: 10 steps
- Parameters: $10 \times 4 \times 5 = 200$

Results:

- Position 0: rank 0/4 (no proof yet)
- Positions 1-4: rank 1/4, **exact solutions**
- Positions 5-9: rank 0/4 (past proof end)
- Training: Closed-form (¡ 0.1 seconds)
- FHE depth: **0**

Remark 4.4 (Automatic Structure Discovery). The rank pattern reveals learned structure:

- Rank 0: No active proof at these positions
- Rank 1: Exact proof step (one rule dominates)
- Wreath product automatically discovered proof length = 4!

No need to specify proof structure—learned from data.

Theorem 4.5 (Exact Inference). *At positions with rank 1, the learned weights yield exact rule prediction:*

$$Pr(\text{predict correct rule}) = 1$$

This is not approximate—it's algebraically exact.

5 Unified Framework

5.1 The Universal Pattern

Theorem 5.1 (When Wreath Products Arise). *A learning problem admits wreath product structure if and only if:*

1. *There exists a group G of local transformations*
2. *There exists a group H acting on positions*
3. *The composition $(f_1, h_1) \cdot (f_2, h_2)$ satisfies:*

$$f_1(p) \cdot f_2(h_1^{-1}(p)) \text{ depends on position } p$$

Proof. ($\Rightarrow$) If problem has wreath structure, positions p index distinct local choices, and permuting positions composes with local transformations non-trivially.

($\Leftarrow$) Position-dependent transformations that interact under permutation define a wreath product by construction (Definition 2.1). □

5.2 Examples

Domain	**Local G**	**Global H**	**Wreath Product**
Rubik's Cube	Corner rotation	Corner permutation	$C_3 \wr S_8$
Cryptography	Block cipher	Block shuffle	Cipher $\wr$ Perm
Fluids	**Fourier mode**	**Spatial grid**	$C_k \wr C_N$
Proofs	**Inference rule**	**Proof step**	**Rules $\wr$ Steps**

5.3 Comparison to Standard Attention

Method	**Parameters**	**Learning**	**FHE Depth**
Softmax attention	$O(n^2)$	Gradient descent	≥ 5
Fixed character	$O(k)$	Closed-form	0
Wreath product	$O(nk)$	**Closed-form**	**0**

Wreath product bridges the gap:

- More expressive than fixed characters (position-dependent)
- Fewer parameters than softmax (algebraic structure)
- Closed-form learning (no iteration)
- FHE-compatible (depth 0)

6 FHE Connection

6.1 Cyclotomic Rings

Theorem 6.1 (Galois Group Connection)**.** *For cyclotomic field $\mathbb{Q}(\zeta_n)$:*

$$Gal(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^*$$

Rotations τ_k are Galois automorphisms $\sigma_k : \zeta_n \mapsto \zeta_n^k$.

Corollary 6.2 (FHE Depth 0)**.** *All wreath product operations achieve FHE depth 0:*

1. *Character projections: rotations (Galois automorphisms)*
2. *Position-dependent weighting: plaintext-ciphertext multiplication*
3. *Summation: depth 0*

Remark 6.3. This is *provably optimal* for FHE: any computation requiring only automorphisms has depth 0.

6.2 Private Computation

Fluid dynamics: Predict weather on encrypted satellite data

Theorem proving: Verify proofs without revealing them (zero-knowledge?)

Both achieve depth 0 $\Rightarrow$ minimal noise growth $\Rightarrow$ practical FHE deployment.

7 Discussion

7.1 Theoretical Implications

Why does algebra structure cognition?

Possible explanations:

1. **Symmetry:** Truth preserved under transformations
2. **Compositionality:** Operations compose algebraically
3. **Universality:** Wreath products universal for extensions (Kaloujnine-Krasner)

Deeper answer: Mathematics discovered groups, characters, and wreath products because these *are* the structures of reasoning.

7.2 Limitations

Navier-Stokes:

- Only tested on smooth flows (Taylor-Green vortex)
- Real turbulence requires validation
- Currently 1D slices only (full 2D extension needed)

Theorem proving:

- Only propositional logic (no quantifiers)
- Trained on single proof (need large corpus)
- No generalization test (can it prove new theorems?)

7.3 Future Work

Immediate:

- Real CFD data (cylinder flow, turbulent boundary layers)
- Larger proof corpus (100+ theorems)
- Predicate logic (quantifiers, induction)

Long-term:

- Sheaf theory (local-global reasoning)
- Cohomology interpretation (obstructions to learning)
- Self-play for theorem generation
- FHE deployment (private weather, private proofs)

8 Conclusion

We demonstrate that wreath product groups provide a natural framework for position-dependent learning in algebraic domains. Through applications to fluid dynamics and automated reasoning, we show:

1. **Exact solutions:** Character theory + least squares achieves MSE $< 10^{-30}$ (fluids) and rank-1 exact solutions (proofs)
2. **No iteration:** Closed-form learning eliminates gradient descent entirely
3. **FHE depth 0:** Rotations only $\Rightarrow$ practical encrypted computation
4. **Universal pattern:** Wreath products arise when local structure varies by position but positions are permutable

The mathematics tells us: *If your problem has position-dependent algebraic structure, wreath products are inevitable.*

Key insight: Optimization can be replaced by algebra when the problem has the right symmetry structure.

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